Laptop Setup Guide

Install and check everything before Day 1 (Mon 26 Oct 2026). About 60 to 90 minutes.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Before Day 1

Install every **Required** item below on the laptop you will use in class, then run the check command. Bring the laptop to the Day 1 onboarding session, where the software check is done together. If you cannot install something because of company policy, tell your trainer before Day 1.

Golden rule: never paste an API key into a file you commit, a notebook cell, a prompt or a chat.

1. Browser and Google account (Day 1 and Day 2)

- Install a current Chrome or Edge.
- Sign in to a Google account and open https://colab.research.google.com to check Colab loads.
- Colab runs every notebook in this program, including the free GPU runtime on Day 25.

```
Check: open Colab, create a notebook, run  print('hello')
```

2. Python 3.12 or newer (Day 2)

- Windows: install from python.org and tick 'Add python.exe to PATH'.
- macOS: install from python.org or with Homebrew (brew install python@3.12).
- Create a virtual environment for each project (Day 3 covers this).

```
python --version          # Windows
python3 --version         # macOS / Linux
python -m pip --version
```

3. VS Code and the OpenAI Codex extension (Day 3)

- Install VS Code from code.visualstudio.com.
- Add the Python and Jupyter extensions (published by Microsoft).
- Add the OpenAI Codex extension (published by OpenAI). You sign in with the program's API key in class on Day 3. Use approved models only.
- Full set-up steps: docs/CODEX_GUIDE.md in the course repository.

```
code --version
```

4. Git and a GitHub account (Day 3)

- Install Git from git-scm.com (macOS: xcode-select --install also works).
- Create or use a GitHub account; you need it for Codex and the capstone repository.

```
git --version
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

5. Node.js LTS with npm (Day 8)

- Install the current LTS release from nodejs.org.
- npm comes with Node.js.

```
node --version
npm --version
```

6. Postman (Day 9)

- Install the Postman desktop app and sign in or skip sign-in.

```
Check: send GET https://httpbin.org/get and see status 200
```

7. Docker Desktop (Day 20)

- Install Docker Desktop and start it once.
- Larger companies need a paid Docker subscription. Check licensing with your IT team.
- Windows: enable WSL 2 when the installer asks.

```
docker --version
docker run --rm hello-world
```

8. Optional items

- MongoDB Compass (Day 4): visual tool for MongoDB. Colab uses mongomock, so this is optional.
- Java 17 (Day 16): only to run PySpark on your laptop. Not needed in Colab.
- Hugging Face account (Day 25): some small models need a free account to download.

9. Python packages

Each day installs its own packages in class (for example openai, pytest, pyspark, langchain, langgraph, mcp, transformers). In VS Code, install them inside a virtual environment:

```
python -m venv .venv
# Windows: .venv\Scripts\activate   macOS/Linux: source .venv/bin/activate
pip install -r requirements.txt
```

10. Adding OPENAI_API_KEY

In Google Colab

- Click the key icon (Secrets) in the left sidebar.
- Add a secret named exactly OPENAI_API_KEY and paste the key as its value.
- Switch on 'Notebook access'. Notebooks read it with google.colab.userdata.

In VS Code (.env file)

- Create a file named .env at the project root containing one line: OPENAI_API_KEY=your-key-here
- Add .env to .gitignore before your first commit.
- Load it with python-dotenv, or set the variable in your terminal.

```
# .gitignore
.env

# check it is ignored
git check-ignore .env      # prints .env if ignored
```

Day 1 software check

Item	Check command or action	Done
Browser + Colab	Run print('hello') in Colab	
Python 3.12+	python --version	
VS Code + extensions	code --version; Python, Jupyter, OpenAI Codex installed	
Git	git --version	
Node.js LTS	node --version; npm --version	
Postman	GET request returns 200	
Docker Desktop	docker run --rm hello-world	

No API keys are needed for Days 1 and 2. Keys are issued in class.